

Module 8

End-to-End Project Workflow

Learning objectives

- Bootstrap a project from a template repo
- Run scripted build/test flows
- Capture and analyze logs
- Commit milestones and push for submission

Prerequisites

- Modules 1–7 or equivalent experience

Learning path

Learning Path

Diagram: Learning Path

(render with mmdc when available)

flowchart LR

A[Clone template] --> B[Run scripts]

B --> C[Logs]

C --> D[Git commits]

D --> E[Submit]

E --> G[Done]

Course context

- Full loop: edit → run → inspect → commit → push
- Reproducibility matters for grading and debug
- Pre-push checks catch common mistakes

Core topics

Self-check

```
$ ./scripts/module8.sh --check
```

Terminal: module_check

```
[0;36m===== [0m
[0;36mModule 8: Environment self-check [0m
[0;36m===== [0m

[0;32m[OK] [0m module8 directory exists: /home/yongfu/proj/universal-verification-methodology/
[0;32m[OK] [0m template/sample_project exists
[0;32m[OK] [0m sample_project/scripts/run_demo.sh exists
[0;32m[OK] [0m logs/sample.log exists

[0;32m[OK] [0m All required checks passed.
```

Run with log capture

```
./scripts/run_demo.sh > logs/run.log 2>&1  
grep -i error logs/run.log
```


Scripted workflow

```
$ cd module8/examples/template/sample_project && chmod u+x  
scripts/run_demo.sh && ./scripts/run_demo.sh
```

Terminal: sample_project_run

```
[2026-05-28T16:40:12+08:00] Run started  
[INFO] Project root: /home/yongfu/proj/universal-verification-methodology/learn_unix_git/module8  
[INFO] Listing src/ and tb/  
/home/yongfu/proj/universal-verification-methodology/learn_unix_git/module8/examples/template/sa  
total 0  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 .  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 ..  
-rwxrwxrwx 1 yongfu yongfu  0 May 28 12:03 .gitkeep  
  
/home/yongfu/proj/universal-verification-methodology/learn_unix_git/module8/examples/template/sa  
total 0  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 .  
drwxrwxrwx 1 yongfu yongfu 4096 May 28 12:03 ..  
-rwxrwxrwx 1 yongfu yongfu  0 May 28 12:03 .gitkeep  
[2026-05-28T16:40:12+08:00] Run finished
```

Log inspection

```
$ grep INFO module8/examples/template/sample_project/logs/run.log  
2>/dev/null | head -3
```

Terminal: grep_log

```
[INFO] Project root: /mnt/d/proj/universal-verification-methodology/learn_unix_git/module8/exasp  
[INFO] Listing src/ and tb/  
[INFO] Project root: /mnt/d/proj/universal-verification-methodology/learn_unix_git/module8/exasp
```

Practice

Exercises

- Walk through `sample_project/`
- Make a commit after a script run
- Review pre-push checklist in `module8`

Summary & next steps

- Unix + Git + scripts together
- You are ready for design courses
- Course complete — keep practicing
- Next: Digital design & verification courses